

MC10212

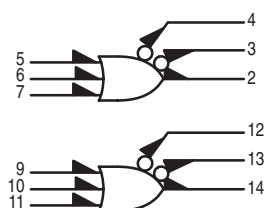
High Speed Dual 3-Input/ 3-Output OR/NOR Gate

The MC10212 is designed to drive up to six transmission lines simultaneously. The multiple outputs of this device also allow the wire "OR"-ing of several levels of gating for minimization of gate and package count.

The ability to control three parallel lines with minimum propagation delay from a single point makes the MC10212 particularly useful in clock distribution applications where minimum clock skew is desired.

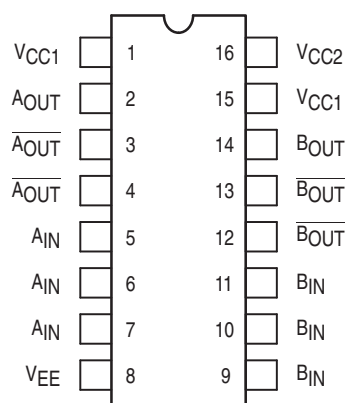
- $P_D = 160 \text{ mW typ/pkg (No Load)}$
- $t_{pd} = 1.5 \text{ ns typ (All Outputs Loaded)}$
- $t_r, t_f = 1.5 \text{ ns typ (20\%–80\%)}$

LOGIC DIAGRAM



$V_{CC1} = \text{PIN 1, 15}$
 $V_{CC2} = \text{PIN 16}$
 $V_{EE} = \text{PIN 8}$

DIP PIN ASSIGNMENT



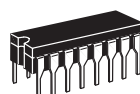
Pin assignment is for Dual-in-Line Package.
 For PLCC pin assignment, see the Pin Conversion Tables on page 18.



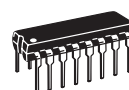
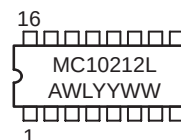
ON Semiconductor

<http://onsemi.com>

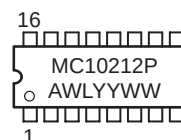
MARKING DIAGRAMS



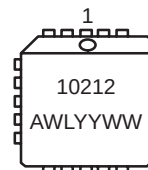
CDIP-16
L SUFFIX
CASE 620



PDIP-16
P SUFFIX
CASE 648



PLCC-20
FN SUFFIX
CASE 775



A = Assembly Location
 WL = Wafer Lot
 YY = Year
 WW = Work Week

ORDERING INFORMATION

Device	Package	Shipping
MC10212L	CDIP-16	25 Units / Rail
MC10212P	PDIP-16	25 Units / Rail
MC10212FN	PLCC-20	46 Units / Rail

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ELECTRICAL CHARACTERISTICS

Characteristic	Symbol	Pin Under Test	Test Limits							Unit
			−30°C		+25°C			+85°C		
			Min	Max	Min	Typ	Max	Min	Max	
Power Supply Drain Current	I _E	8		42		30	38		42	mAdc
Input Current	I _{inH}	5, 6, 7		650			410		410	μAdc
	I _{inL}	5, 6, 7	0.5		0.5			0.3		μAdc
Output Voltage Logic 1	V _{OH}	2 3 4	−1.060 −1.060 −1.060	−0.890 −0.890 −0.890	−0.960 −0.960 −0.960		−0.810 −0.810 −0.810	−0.890 −0.890 −0.890	−0.700 −0.700 −0.700	Vdc
Output Voltage Logic 0	V _{OL}	2 3 4	−1.890 −1.890 −1.890	−1.675 −1.675 −1.675	−1.850 −1.850 −1.850		−1.650 −1.650 −1.650	−1.825 −1.825 −1.825	−1.615 −1.615 −1.615	Vdc
Threshold Voltage Logic 1	V _{OHA}	2 3 4	−1.080 −1.080 −1.080		−0.980 −0.980 −0.980			−0.910 −0.910 −0.910		Vdc
Threshold Voltage Logic 0	V _{OLA}	2 3 4		−1.655 −1.655 −1.655			−1.630 −1.630 −1.630		−1.595 −1.595 −1.595	Vdc
Switching Times (50Ω Load)										ns
Propagation Delay	t ₅₊₂₊	2	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t _{5−2−}	2	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t _{5+3−}	3	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t _{5−3+}	3	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t _{5+4−}	4	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t _{5−4+}	4	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
Rise Time (20 to 80%)	t ₂₊	2	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t ₃₊	3	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t ₄₊	4	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
Fall Time (20 to 80%)	t _{2−}	2	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t _{3−}	3	1.0	2.6	1.0	1.5	2.5	1.0	2.8	
	t _{4−}	4	1.0	2.6	1.0	1.5	2.5	1.0	2.8	

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ELECTRICAL CHARACTERISTICS (continued)

@ Test Temperature			TEST VOLTAGE VALUES (Volts)					(V _{CC}) Gnd
			V _{IHmax}	V _{ILmin}	V _{IHAmin}	V _{ILAmax}	V _{EE}	
			–30°C	–0.890	–1.890	–1.205	–1.500	–5.2
			+25°C	–0.810	–1.850	–1.105	–1.475	–5.2
			+85°C	–0.700	–1.825	–1.035	–1.440	–5.2
Characteristic	Symbol	Pin Under Test	TEST VOLTAGE APPLIED TO PINS LISTED BELOW					(V _{CC}) Gnd
			V _{IHmax}	V _{ILmin}	V _{IHAmin}	V _{ILAmax}	V _{EE}	
Power Supply Drain Current	I _E	8					8	1, 15, 16
Input Current	I _{inH}	5, 6, 7	5, 6, 7*				8	1, 15, 16
	I _{inL}	5, 6, 7		5, 6, 7*			8	1, 15, 16
Output Voltage Logic 1	V _{OH}	2	5				8	1, 15, 16
		3					8	1, 15, 16
		4					8	1, 15, 16
Output Voltage Logic 0	V _{OL}	2					8	1, 15, 16
		3	5				8	1, 15, 16
		4	5				8	1, 15, 16
Threshold Voltage Logic 1	V _{OHA}	2			5		8	1, 15, 16
		3				5	8	1, 15, 16
		4				5	8	1, 15, 16
Threshold Voltage Logic 0	V _{OLA}	2				5	8	1, 15, 16
		3			5		8	1, 15, 16
		4			5		8	1, 15, 16
Switching Times (50Ω Load)					Pulse In	Pulse Out	–3.2 V	+2.0 V
Propagation Delay	t ₅₊₂₊	2			5	2	8	1, 15, 16
	t _{5–2–}	2			5	2	8	1, 15, 16
	t _{5+3–}	3			5	3	8	1, 15, 16
	t _{5–3+}	3			5	3	8	1, 15, 16
	t _{5+4–}	4			5	4	8	1, 15, 16
	t _{5–4+}	4			5	4	8	1, 15, 16
Rise Time (20 to 80%)	t ₂₊	2			5	2	8	1, 15, 16
	t ₃₊	3			5	3	8	1, 15, 16
	t ₄₊	4			5	4	8	1, 15, 16
Fall Time (20 to 80%)	t _{2–}	2			5	2	8	1, 15, 16
	t _{3–}	3			5	3	8	1, 15, 16
	t _{4–}	4			5	4	8	1, 15, 16

* Individually test each input using the pin connections shown.

Each MECL 10,000 series circuit has been designed to meet the dc specifications shown in the test table, after thermal equilibrium has been established. The circuit is in a test socket or mounted on a printed circuit board and transverse air flow greater than 500 linear fpm is maintained. Outputs are terminated through a 50-ohm resistor to –2.0 volts. Test procedures are shown for only one gate. The other gates are tested in the same manner.